- DEF. VON NEUMAN ENTROPY

$$S(\rho) := -\mathrm{tr}(\rho \log \rho) \quad \text{where } \rho \text{ is a density matrix}$$

- $$\rho = \sum_{i=1}^{d} p_i |v_i\rangle\langle v_i| \implies S(\rho) = -\sum_{i=1}^{d} p_i \log(p_i) = H(\{p_i\})$$

  ($\rho = \sum_{i=1}^{d} p_i |v_i\rangle\langle v_i|$: EIGENDEC.; p_i: eigenvalues (with $\sum_{i=1}^{d} p_i = 1$); $H(\{p_i\})$: SHANNON ENTROPY)

PROOF:

In general we have $f(\rho) = \sum_{i=1}^{d} f(p_i) |v_i\rangle\langle v_i|$.

FACT ⊛:

This because every $f(\cdot)$ can be written as $f(\rho) = \sum_{j=0}^{\infty} c_j \rho^j$

$$\Rightarrow f(\rho) = \sum_{j=0}^{\infty} c_j \rho^j \underbrace{\sum_{i=1}^{d} |v_i\rangle\langle v_i|}_{=\mathbb{1}} = \sum_i \sum_j c_j \underbrace{\left(\rho^j |v_i\rangle\right)}_{p_i^j |v_i\rangle} \langle v_i| =$$

$$= \sum_i \left(\sum_j c_j p_i^j\right) |v_i\rangle\langle v_i| = \sum_{i=1}^{d} f(p_i) |v_i\rangle\langle v_i|$$

($\sum_j c_j p_i^j = f(p_i)$)

- So given $f(\rho) := \rho \log(\rho) \Rightarrow \rho \log(\rho) = \sum_{i=1}^{d} (p_i \log(p_i)) |v_i\rangle\langle v_i|$

$$\Rightarrow S(\rho) := -\mathrm{tr}(\rho \log \rho) = -\sum_{i=1}^{d} (p_i \log(p_i)) \underbrace{\mathrm{tr}(|v_i\rangle\langle v_i|)}_{=1}$$

$$= -\sum_{i=1}^{d} p_i \log p_i$$

OBS.1

$$0 \leq S(\rho) \leq \log(d)$$

PROOF:

$S(\rho) = H(\{p_i\})$ and we showed that $0 \leq H(x) \leq \log(d)$.

OBS!

- $S(\rho) = 0 \iff \rho = |\psi\rangle\langle\psi|$ (PURE)
- $S(\rho) = \log(d) \iff \rho = \frac{\mathbb{1}}{d}$ (MAXIMALLY MIXED STATE)

PROOF:

- $S(\rho) = H(\{p_i\}) = 0 \iff \{p_i\}$ is a peaked distribution i.e. $p_i = \delta_{i,a}$ $\forall i$.
  (↑ PREVIOUS LECTURE)
- $S(\rho) = H(\{p_i\}) = \log(d) \iff \{p_i\}$ is a flat distribution i.e. $p_i = \frac{1}{d}$ $\forall i$.
  $\iff \rho = \sum_{i=1}^{d} p_i |v_i\rangle\langle v_i| = \frac{1}{d} \underbrace{\sum_{i=1}^{d} |v_i\rangle\langle v_i|}_{\mathbb{1}} \Rightarrow \mathbb{1}$

- $S(\rho) = S(U\rho U^\dagger)$ $\forall$ U unitary.

PROOF:

$\rho = \sum_i p_i |v_i\rangle\langle v_i|$ and $U\rho U^\dagger = \sum_i p_i (U|v_i\rangle)(\langle v_i|U^\dagger)$ have the same eigenvalues and $S(\cdot)$ depends only by the eigenvalues.

- DEF: (RELATIVE ENTROPY)

$$S(\rho_1 \| \rho_2) := \mathrm{tr}[\rho_1 \log(\rho_1)] - \mathrm{tr}[\rho_1 \log(\rho_2)]$$
$$= -S(\rho_1) - \mathrm{tr}[\rho_1 \log(\rho_2)]$$

TEO 8: (KLEIN INEQUALITY)

$$S(\rho_1\|\rho_2) \geq 0 \qquad \left(S(\rho_1\|\rho_2)=0 \iff \rho_1=\rho_2\right)$$

PROOF:

$$\rho_1 = \sum_{i=1}^{d} p_i |v_i\rangle\langle v_i| \quad , \quad \rho_2 = \sum_{j=1}^{d} q_j |w_j\rangle\langle w_j|$$

$$S(\rho_1\|\rho_2) = -S(\rho_1) - \mathrm{tr}\left[\rho_1 \log(\rho_2)\right]$$

$$= -S(\rho_1) - \mathrm{tr}\left[\sum_{i=1}^{d} p_i |v_i\rangle\langle v_i| \log(\rho_2)\right]$$

$$= -S(\rho_1) - \sum_{i=1}^{d} p_i \langle v_i| \log(\rho_2) |v_i\rangle$$

$$= -S(\rho_1) - \sum_{i=1}^{d}\sum_{j=1}^{d} p_i \log(q_j) \underbrace{\langle v_i|w_j\rangle\langle w_j|v_i\rangle}_{|\langle v_i|w_j\rangle|^2} =$$

$$\log(\rho_2) = \sum_{j=1}^{d} \log(q_j) |w_j\rangle\langle w_j| \quad (\star)$$

$$= -S(\rho_1) - \sum_{i=1}^{d}\sum_{j=1}^{d} p_i \log(q_j) |\langle v_i|w_j\rangle|^2$$

$$= +\sum_{i=1}^{d} p_i \log(p_i) - \sum_{i=1}^{d}\sum_{j=1}^{d} p_i \log(q_j) |\langle v_i|w_j\rangle|^2$$

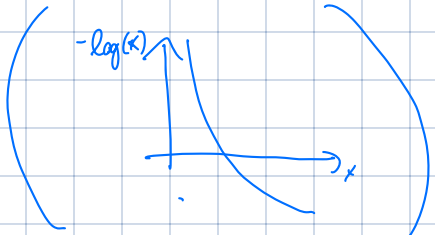
$$= +\sum_{i}\sum_{j} |\langle v_i|w_j\rangle|^2 p_i \log(p_i) - \sum_{i=1}^{d}\sum_{j} |\langle v_i|w_j\rangle|^2 p_i \log(q_j)$$

$$\sum_{j} |\langle v_i|w_j\rangle|^2 = 1$$

$$= +\sum_{i}\sum_{j} |\langle v_i|w_j\rangle|^2 p_i \left(\log\left(\frac{p_i}{q_j}\right)\right) = \sum_{i}\sum_{j} \underbrace{|\langle v_i|w_j\rangle|^2}_{P(i,j)} p_i \left(-\log\left(\frac{q_j}{p_i}\right)\right)$$

$$\geq \sum_i p_i \left[-\log\left(\sum_j P(i,j) \frac{q_j}{p_i} \right) \right] = \sum_i p_i \left[-\log\left(\frac{r_i}{p_i} \right) \right]$$

- $\sum_j P(i,j) = 1$, $P(i,j) \geq 0$
- $-\log(x)$ CONVEX



- $r_i := \sum_j P(i,j)\, q_j$

Probab. distr.

($\sum_i r_i = 1$, $r_i \geq 0$)

$$= \sum_i p_i \log\left(\frac{p_i}{r_i} \right) \geq 0$$

(previous lecture)

- The first inequality is saturated when $p_i = q_j \quad \forall i,j$

$\Downarrow$

$p_i = q_j = \frac{1}{d} \quad \forall i,j$

or when $P(i,j) = \delta_{j,i_0} \quad \forall j = 1, \dots, d$

- But $P(i,j) = |\langle v_i | w_j \rangle|^2 = \delta_{j,i_0} \quad \forall j = 1, \dots, d \iff |v_i\rangle = |w_{i_0}\rangle \quad \forall i$

$\uparrow$

$$|v_i\rangle = \sum_j \langle w_j | v_i \rangle |w_j\rangle$$

$$\langle w_j | v_i \rangle = \delta_{j,i_0} e^{-i\phi(i_0)}$$

$$= e^{i\phi(i_0)} |w_{i_0}\rangle$$

- The second inequality is saturated when $p_i = r_i \quad \forall i$ (previous lecture)

$$\iff p_i = \sum_j P(i,j) q_j = \sum_j \delta_{j,i_0} q_j = q_{i_0} \quad \forall i$$

$$\iff \rho_1 = \sum_i p_i |v_i\rangle\langle v_i| = \sum_i q_{i_0} |w_{i_0}\rangle\langle w_{i_0}| = \rho_2$$

- OBS.
  - $S(\rho_1 \| \rho_2) \neq S(\rho_2 \| \rho_1)$ (NOT SYMMETRIC)
  - $\text{supp}(\rho_1) \cap \text{Ker}(\rho_2) \neq \{\vec{0}\} \Rightarrow S(\rho_1 \| \rho_2) = +\infty$

PROOF:

$$S(\rho_1 \| \rho_2) = -S(\rho_1) - \sum_{i=1}^{d} \sum_{j=1}^{d} p_i \log(q_j) |\langle v_i | w_j \rangle|^2$$

If $q_j = 0$ but $p_i |\langle v_i | w_j \rangle| \neq 0 \ \forall i \Rightarrow S(\rho_1 \| \rho_2) = +\infty$

$|w_j\rangle \in \text{Ker}(\rho_2)$ $\Leftrightarrow$ $|w_j\rangle \notin \text{Ker}^{\perp}(\rho_1) = \text{supp}(\rho_1)$

- So $\text{Ker}(\rho_2) \cap \text{supp}(\rho_1) \neq \{0\} \Rightarrow S(\rho_1 \| \rho_2) = +\infty \Rightarrow$!!

- $O = O_1 \otimes O_2$ with $O_1 = O_1^{\dagger}$, $O_2 = O_2^{\dagger}$

$$\log(O) = \log(O_1) \otimes \mathbb{1} + \mathbb{1} \otimes \log(O_2) = (\star)$$

PROOF:

$$\log(O) = \log(O_1 \otimes O_2) \overset{\uparrow}{=} \log\left[\sum_{i,j} \lambda_i^{(1)} \lambda_j^{(2)} \left(|v_i^{(1)}\rangle\langle v_i^{(1)}| \otimes |v_i^{(2)}\rangle\langle v_i^{(2)}|\right)\right] =$$

$O_1 = \sum_i \lambda_i^{(1)} |v_i^{(1)}\rangle\langle v_i^{(1)}|$, $O_2 = \sum_j \lambda_j^{(2)} |v_j^{(2)}\rangle\langle v_j^{(2)}|$

$\uparrow$ eigenvectors for $O_1 \otimes O_2$

$$= \sum_{i,j} \log\left(\lambda_i^{(1)} \lambda_j^{(2)}\right) \left(|v_i^{(1)}\rangle\langle v_i^{(1)}| \otimes |v_i^{(2)}\rangle\langle v_i^{(2)}|\right) =$$

$f(A) = \sum_i f(\lambda_i) |v_i\rangle\langle v_i|$ where $A = \sum_i \lambda_i |v_i\rangle\langle v_i|$

$\log(ab) = \log(a) + \log(b)$

$$= \sum_{i,j} \log\left(\lambda_i^{(1)}\right) \left(|v_i^{(1)}\rangle\langle v_i^{(1)}| \otimes |v_i^{(2)}\rangle\langle v_i^{(2)}|\right) + \sum_{i,j} \log\left(\lambda_j^{(2)}\right) \left(|v_i^{(1)}\rangle\langle v_i^{(1)}| \otimes |v_i^{(2)}\rangle\langle v_i^{(2)}|\right)$$

$$= \sum_i \log\left(\lambda_i^{(1)}\right) \left(|v_i^{(1)}\rangle\langle v_i^{(1)}| \otimes \mathbb{1}\right) + \sum_j \log\left(\lambda_j^{(2)}\right) \left(\mathbb{1} \otimes |v_i^{(2)}\rangle\langle v_i^{(2)}|\right) =$$

$$= \log(O_1) \otimes \mathbb{1} + \log(O_2) \otimes \mathbb{1}$$

COR.

- $O = O_1 \otimes O_2 \otimes \cdots \otimes O_n \quad , O_i = O_i^\dagger \quad \forall i$
- $\log(O) = \sum_{i=1}^{N} \mathbb{1}_{1,\dots,i-1} \otimes \log(O_i) \otimes \mathbb{1}_{i+1,\dots n}$

COR:

$$S(P_1 \otimes P_2 \otimes \cdots \otimes P_N) = \sum_{i=1}^{N} S(P_i)$$

PROOF:

$$\log(\rho) = \sum_{i=1}^{N} \mathbb{1}_{1,\dots,i-1} \otimes \log(\rho_i) \otimes \mathbb{1}_{i+1,\dots n}$$

$$S(\rho) = -\mathrm{tr}\left(\rho \log(\rho)\right) = -\sum_{i=1}^{N} \mathrm{tr}\left(\rho \left(\mathbb{1}_{1,\dots,i-1} \otimes \log(\rho_i) \otimes \mathbb{1}_{i+1,\dots n}\right)\right) =$$

$$= -\sum_{i=1}^{N} \mathrm{tr}\left(\underbrace{\mathrm{tr}_{\hat{i}}(\rho)}_{\rho_i} \log(\rho_i)\right) = \sum_{i=1}^{N} S(\rho_i)$$

THEOR. ②

- $\rho_A = \mathrm{tr}_B(\rho_{AB}) \quad , \rho_B = \mathrm{tr}_A(\rho_{AB})$
- $S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B) \quad \left(S(\rho_{AB}) = S(\rho_A) + S(\rho_B) \iff \rho_{AB} = \rho_A \otimes \rho_B\right)$

PROOF:

- $S(\rho_{AB} \| \rho_A \otimes \rho_B) \geq 0$

  ↑ KLEIN INEQUALITY

- $\underbrace{S(\rho_{AB} \| \rho_A \otimes \rho_B)}_{\geq 0} = -S(\rho_{AB}) - \mathrm{tr}\left(\rho_{AB} \underbrace{\log(\rho_A \otimes \rho_B)}_{\mathbb{1} \leftarrow \otimes}\right)$

  $\log(\rho_A) \otimes \mathbb{1} + \mathbb{1} \otimes \log(\rho_B)$

$$= -S(\rho_{AB}) - \underbrace{tr\left(\rho_{AB}\log(\rho_A)\otimes\mathbb{1}\right)}_{tr(\rho_A\log(\rho_A))} - tr\left(\rho_{AB}\,\mathbb{1}\otimes\log(\rho_B)\right)$$

$$= -S(\rho_{AB}) + S(\rho_A) + S(\rho_B) \geq 0 \quad \Rightarrow \boxtimes$$

- "$=$" $\Leftrightarrow$ $S(\rho_{AB}\,\|\,\rho_A\otimes\rho_B) = 0 \Leftrightarrow \rho_{AB} = \rho_A\otimes\rho_B$

FACT ③ :

- $\rho_A := tr_B(\rho)$, $\rho_B := tr_A(\rho)$
- $\rho = |\psi\rangle\langle\psi|$ pure $\Rightarrow S(\rho_A) = S(\rho_B)$

(This is not true in general if ρ is not pure. Find a counter-example ⑤.)

PROOF:

SCHMIDT. $|\psi\rangle = \sum_{i=1}^{\min(d_A,d_B)} \sqrt{\lambda_i}\,|v_i^A\rangle\otimes|v_i^B\rangle$ with $\langle v_i^A, v_j^A\rangle = \delta_{i,j}$, $\langle v_i^B, v_j^B\rangle = \delta_{i,j}$.

SUBPROOF:

- $|\psi\rangle = \sum_{i,j}\langle i,j|\psi\rangle\,|i,j\rangle = \sum_{i,j}(UDV^\dagger)_{i,j}\,|i,j\rangle =$

  ($\langle i,j|\psi\rangle = (C)_{i,j}$; SVD: $C = UDV^\dagger$)

$$= \sum_{\ell}\sum_{i,j} U_{i,\ell}\,D_{\ell\ell}\,(V^\dagger)_{\ell,j}\,|i\rangle\otimes|j\rangle = \sum_{\ell}\lambda_\ell\left(\sum_i U_{i\ell}|i\rangle\right)\otimes\left[\sum_j (V^\dagger)_{\ell,j}|j\rangle\right]$$

($\langle i|U|\ell\rangle = U_{i\ell}$; $\sqrt{\lambda_\ell} := D_{\ell\ell} \geq 0$; $U|\ell\rangle = |v_i^A\rangle$; $V^\dagger|\ell\rangle = |v_i^B\rangle$)

$$tr_B(|\psi\rangle\langle\psi|) = \sum_i \lambda_i\,|v_i^{(A)}\rangle\langle v_i^{(A)}|$$

$$tr_A(|\psi\rangle\langle\psi|) = \sum_i \lambda_i\,|v_i^{(B)}\rangle\langle v_i^{(B)}|$$

$\Rightarrow$ Same eigenvalues $\Rightarrow$ same entropies.

THEOREM ④

$$S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$$

PROOF:

- Given a mixed state ρ_{AB}, then $\exists$ R such that $\rho_{AB} = \mathrm{tr}_R[\underbrace{|\psi\rangle\langle\psi|}_{\rho_{ABR}}]$ with $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_R$

SUBPROOF: (ALREADY SEEN)

E.g. take $|\psi\rangle := \sqrt{\rho_{AB}} \otimes \mathbb{1}_R \left(\sum_{i=1}^{d} |i\rangle_{AB} \otimes |i\rangle_R \right)$ and verify.



- We have:

$$S(\rho_B) = S(\rho_{AR}) \leq S(\rho_A) + S(\rho_R) = S(\rho_A) + S(\rho_{AB})$$

($S(\rho_B) = S(\rho_{AR})$: • FACT ③ • ρ_{ABR} pure; $S(\rho_{AR}) \leq S(\rho_A) + S(\rho_R)$: TH ②; $S(\rho_A) + S(\rho_R) = S(\rho_A) + S(\rho_{AB})$: • $S(\rho_{AB}) = S(\rho_R)$ • FACT ③ • ρ_{ABR} pure)

$$\Rightarrow \boxed{S(\rho_{AB}) \geq S(\rho_B) - S(\rho_A)} \equiv \star_1$$

- $$S(\rho_A) = S(\rho_{BR}) \leq S(\rho_B) + S(\rho_R) = S(\rho_B) + S(\rho_{AB})$$

($S(\rho_A) = S(\rho_{BR})$: FACT ③; $S(\rho_{BR}) \leq S(\rho_B) + S(\rho_R)$: TH ②; $S(\rho_B) + S(\rho_R) = S(\rho_B) + S(\rho_{AB})$: FACT ③)

$$\Rightarrow \boxed{S(\rho_{AB}) \geq S(\rho_A) - S(\rho_B)} \equiv \star_2$$

$\Rightarrow \square$ ($\star_1, \star_2$)

OBS] Also in the classical case I have $H(X,Y) \geq |H(X) - H(Y)|$

but we did not mention it because in the classical case we have an even stronger inequality: $H(X,Y) \geq \text{MAX}(H(X), H(Y))$ which can be violated quantumly.

FALSE NO! FALSE!

$$S(\rho_{AB}) \geq \max(S(\rho_A), S(\rho_B))$$

NO! NO!

PROOF.

COUNTER-EXAMPLE: $\rho_{AB} = |\psi\rangle\langle\psi|$ with $|\psi\rangle = |\Omega\rangle$ max. entangled state.

- $S(\rho_{AB}) = 0$ ← state pure.
- $\max(S(\rho_A), S(\rho_B)) = \max\left(S\left(\frac{\mathbb{1}}{d_A}\right), S\left(\frac{\mathbb{1}}{d_B}\right)\right) \neq 0$ (each $\neq 0$)

DEF.

ρ Separable $\iff$ $\rho = \sum_{i=1}^{N} p_i \rho_1^{(i)} \otimes \rho_2^{(i)}$ with $\begin{cases} p_i \geq 0 \\ \sum_{i=1}^{N} p_i = 1 \end{cases}$

$\rho_1^{(i)}$ and $\rho_2^{(i)}$ states.

DEF

ρ entangled $\iff$ ρ Not separable.

FACT:

ρ separable $\Rightarrow$ $S(\rho_{AB}) \geq S(\rho_A)$

PROOF: NOT SHOWN HERE.

THEOR. (CONCAVITY)

$$S\left(\sum_{i=1}^{N} p_i \rho_i\right) \geq \sum_{i=1}^{N} p_i S(\rho_i) \qquad \left(S\left(\sum_{i=1}^{N} p_i \rho_i\right) = \sum_{i=1}^{N} p_i S(\rho_i) \Leftrightarrow \rho_i = \rho \ \forall i\right)$$

PROOF:

- $\rho_{AB} := \sum_{i=1}^{N} p_i \rho_i \otimes |i\rangle\langle i| \underset{\text{EIGEN.}}{=} \sum_{i=1}^{N} \sum_{j=1}^{d} p_i \lambda_j^{(i)} |v_j^{(i)}\rangle\langle v_j^{(i)}| \otimes |i\rangle\langle i|$

$$S(\rho_{AB}) = H\left(\{p_i \lambda_j^{(i)}\}\right) = -\sum_{i,j} p_i \lambda_j^{(i)} \log\left(p_i \lambda_j^{(i)}\right) = -\sum_{i,j} p_i \lambda_j^{(i)} \log(p_i) - \sum_{i,j} p_i \lambda_j^{(i)} \log\left(\lambda_j^{(i)}\right)$$

$$= H(\{p_i\}) + \sum_{i=1}^{N} p_i S(\rho_i)$$

$\left(\text{since } \sum_j \lambda_j^{(i)} = 1\right)$

- $\rho_A = \mathrm{tr}_B(\rho_{AB}) = \sum_{i=1}^{N} p_i \rho_i$

$$S(\rho_A) = S\left(\sum_{i=1}^{N} p_i \rho_i\right)$$

- $\rho_B = \mathrm{tr}_A(\rho_{AB}) = \sum_{i=1}^{N} p_i |i\rangle\langle i|$

$$S(\rho_B) = H(\{p_i\})$$

- $S(\rho_{AB}) \underset{\text{SUB-ADD.}}{\leq} S(\rho_A) + S(\rho_B)$

$$\| \qquad\qquad \| \qquad\qquad \| \qquad\qquad \Rightarrow \square$$

$$H(\{p_i\}) + \sum_{i=1}^{N} p_i S(\rho_i) \leq S\left(\sum_{i=1}^{N} p_i \rho_i\right) + H(\{p_i\})$$

THEOR.

$$\sum_{i=1}^{N} p_i S(\rho_i) \leq S\left(\sum_{i=1}^{N} p_i \rho_i\right) \leq H(\{p_i\}) + \sum_{i=1}^{N} p_i S(\rho_i)$$

PROOF:

- $S\left(\sum_{i=1}^{N} p_i \rho_i\right) = S\left(\sum_{i=1}^{N}\sum_{s=1}^{d} p_i \lambda_s^{(i)} |v_s^{(i)}\rangle\langle v_s^{(i)}|\right)$

  $\uparrow$ $\rho_i = \sum_{s=1}^{d} \lambda_s |v_s\rangle\langle v_s|$

- $|\psi\rangle_{AB} := \sum_{i=1}^{N}\sum_{s=1}^{d} \sqrt{p_i \lambda_s^{(i)}}\, |v_s^{(i)}\rangle \otimes |(i,s)\rangle$ (orthonorm. basis)

- $S(\rho_A) = S\left(\mathrm{tr}_B[|\psi\rangle\langle\psi|]\right) = S\left(\mathrm{tr}_A[|\psi\rangle\langle\psi|]\right) = S(\rho_B)$

  $|\psi\rangle\langle\psi|$ pure

  $S\left(\sum_{i,s} p_i \lambda_s^{(i)} |v_s^{(i)}\rangle\langle v_s^{(i)}|\right)$

  $=$

  $S\left(\sum_{i=1}^{N} p_i \rho_i\right)$

$$\Rightarrow S(\rho_B) = S\left(\sum_{i=1}^{N} p_i \rho_i\right)$$

- $\rho_B = \mathrm{tr}_A[|\psi\rangle\langle\psi|] = \mathrm{tr}_A\left(\sum_{i=1}^{N}\sum_{s=1}^{d}\sum_{k=1}^{N}\sum_{l=1}^{d} \sqrt{p_i \lambda_s^{(i)}}\sqrt{p_k \lambda_l^{(k)}}\, |v_s^{(i)}\rangle\langle v_l^{(k)}| \otimes |(i,s)\rangle\langle(k,l)|\right) =$

$$= \sum_{m=1}^{d} \langle m| \otimes \mathbb{1}\left(\sum_{i=1}^{N}\sum_{s=1}^{d}\sum_{k=1}^{N}\sum_{l=1}^{d} \sqrt{p_i \lambda_s^{(i)}}\sqrt{p_k \lambda_l^{(k)}}\, |v_s^{(i)}\rangle\langle v_l^{(k)}| \otimes |(i,s)\rangle\langle(k,l)|\right) |m\rangle \otimes \mathbb{1}$$

$$= \sum_{m=1}^{d}\sum_{i=1}^{N}\sum_{s=1}^{d}\sum_{k=1}^{N}\sum_{l=1}^{d} \sqrt{p_i \lambda_s^{(i)}}\sqrt{p_k \lambda_l^{(k)}}\, \langle m|v_s^{(i)}\rangle\langle v_l^{(k)}|m\rangle\, |(i,s)\rangle\langle(k,l)|$$

$$= \sum_{i=1}^{N}\sum_{s=1}^{d}\sum_{k=1}^{N}\sum_{l=1}^{d} \sqrt{p_i \lambda_s^{(i)}}\sqrt{p_k \lambda_l^{(k)}}\, \langle v_l^{(k)}|v_s^{(i)}\rangle\, |(i,s)\rangle\langle(k,l)|$$

- If I perform a measurement on B (with PVM $\{|(i,s)\rangle\langle(i,s)|\}_{i,s}$, i.e. $P_{i,s}$), then the post measurement state will be:

$$\rho_B' = \sum_{i'=1}^{d}\sum_{s'=1}^{N} \mathrm{tr}\left(\rho_B P_{i',s'}\right)\left(\frac{P_{i',s'}\,\rho_B\,P_{i',s'}}{\mathrm{tr}\left(\rho_B P_{i',s'}\right)}\right)$$

$$= \sum_{i'=1}^{d}\sum_{s'=1}^{N} P_{i',s'}\,\rho_B\,P_{i',s'} = \sum_{i',s'} p_{i'} \lambda_{s'}^{(i')}\, |(i',s')\rangle\langle(i',s')|$$

$$\rho_B' = \sum_{i,s} p_i \lambda_s^{(i)} |(i,s)\rangle\langle(i,s)|$$

$$S(\rho_B') = H\left(\{p_i \lambda_s^{(i)}\}\right) = H(\{p_i\}) + \sum_i p_i \underbrace{\sum_{s=1}^{d} \lambda_s^{(i)} \log(\lambda_s^{(i)})}_{S(\rho_i)}$$

- **LEMMA** · $\{P_k\}_k$ PVM set $\Rightarrow S\left(\underbrace{\sum_k P_k \rho P_k}_{\rho'}\right) \geq S(\rho)$

(PROOF LATER)

↑ state after measurement.

- $S(\rho_B') \geq S(\rho_B) = S\left(\sum_i p_i \rho_i\right)$

$\|$

$H(\{p_i\}) + \sum_i p_i S(\rho_i)$

- **LEMMA** · $\{P_k\}_k$ PVM set $\Rightarrow S\left(\underbrace{\sum_k P_k \rho P_k}_{\rho'}\right) \geq S(\rho)$

↑ state after measurement.

SUB PROOF:

- $0 \leq S(\rho \| \rho') = -S(\rho) - \mathrm{tr}(\rho \log(\rho'))$

↑ KLEIN INEQ

- $-\mathrm{tr}(\rho \log(\rho')) \geq S(\rho)$

- $-\operatorname{tr}\left(\rho \log(\rho')\right) = -\operatorname{tr}\left(\rho' \log(\rho')\right) \equiv S(\rho') \Rightarrow \square$

SUB SUB PROOF:

- $\operatorname{tr}\left(\rho \log(\rho')\right) \overset{\left(\text{PVM} \Rightarrow \cdot P_i \cdot P_j = \delta_{i,j} P_i,\ \cdot \sum_i P_i = \mathbb{1}\right)}{=} \operatorname{tr}\left(\sum_i P_i \rho \log(\rho')\right) = \sum_i \operatorname{tr}\left(P_i^2 \rho \log(\rho')\right) \overset{\operatorname{tr}(AB) = \operatorname{tr}(BA)}{=} \sum_i \operatorname{tr}\left(P_i \rho \log(\rho') P_i\right) =$

- $\log(\rho') P_i = P_i \log(\rho')$

- $\rho' = \sum_j P_j \rho P_j \Rightarrow \rho' P_i = P_i \rho P_i = P_i \rho' \Rightarrow \log(\rho') P_i = P_i \log(\rho')$

  $\log(\rho') = \sum_{k=0}^{\infty} c_k \rho^k$

$= \sum_i \operatorname{tr}\left(P_i \rho P_i \log(\rho')\right) = \sum_i \operatorname{tr}\left(\rho' \log(\rho')\right) \Rightarrow \square$

THEOR. (SSA) (STRONG SUB-ADDITIVITY)

$$S(\rho_{ABC}) + S(\rho_C) \leq S(\rho_{AC}) + S(\rho_{BC})$$

with $\rho_{AB} = \operatorname{tr}_C(\rho_{ABC})$, $\rho_{BC} = \operatorname{tr}_A(\rho_{ABC})$, $\rho_A = \operatorname{tr}_{BC}(\rho_{ABC})$, ---

OBS STRONG SUB-ADDITIVITY $\Rightarrow$ SUB-ADDITIVITY

PROOF:

- $\rho_{ABC} := \rho_{AB} \otimes |0\rangle\langle 0|$

$$\underbrace{S(\rho_{ABC})}_{\text{PRODUCT STATE} \Rightarrow \; S(\rho_{AB}) + \underbrace{S(|0\rangle\langle 0|)}_{=0 \Leftarrow \text{PURE}}} + \underbrace{S(\rho_C)}_{S(|0\rangle\langle 0|) = 0} \leq \underbrace{S(\rho_{AC})}_{S(\rho_A \otimes |0\rangle\langle 0|) = S(\rho_A) + \underbrace{S(|0\rangle\langle 0|)}_{=0}} + \underbrace{S(\rho_{BC})}_{S(\rho_B \otimes |0\rangle\langle 0|) = S(\rho_B) + \underbrace{S(|0\rangle\langle 0|)}_{=0}}$$

$\Rightarrow S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B) \Rightarrow \square$

PROP.

- $\rho^{\otimes n} := \overbrace{\rho \otimes \cdots \otimes \rho}^{n\text{-times}}$
- $S(\rho^{\otimes n}) \overset{\text{SUB-ADD. (PRODUCT)}}{\leq} n S(\rho)$
- $S(\rho^{\otimes n} \| \sigma^{\otimes n}) = n S(\rho \| \sigma)$

PROOF:

$$S(\rho^{\otimes n} \| \sigma^{\otimes n}) = -S(\rho^{\otimes n}) + \mathrm{tr}(\rho^{\otimes n} \log(\sigma^{\otimes n})) =$$

$$= -n S(\rho) + n \,\mathrm{tr}(\rho \log(\sigma)) = n S(\rho \| \sigma).$$

($\uparrow$ · $S(\rho^{\otimes n}) = n S(\rho)$ · $\log(A \otimes B) = \log A \otimes \mathbb{1} + \mathbb{1} \otimes \log(B)$)